

Author: Trial project submission

Date: August 2026

One-page summary for PDF export

Problem & goal

Breast cancer risk calculators return a percentage that many users find either meaningless or frighteningâ neither reaction reliably drives appropriate follow-up. This app adds a 2â 3 minute conversational layer: explain the number in plain language, branch on elevated vs average risk, and motivate concrete next steps without diagnosing or impersonating a clinician.

Theories chosen & operationalization

1. Health Belief Model (HBM) â Rosenstock, 1974; updated in breast screening literature

Why: HBM predicts preventive action when people perceive susceptibility and severity *accurately*, believe benefits outweigh barriers, and feel capable of acting.

In the dialogue (shared/dialogue.ts):

- Elevated branch: Calibrate severity (â most women with this estimate do *not* get cancerâ); cue to action (clinician visit in 2â 4 weeks); surface barriers (cost, false alarms); build self-efficacy (â youâ re already taking a helpful stepâ).
- Average branch: Reassure without false certainty; cue routine mammography; affirm existing healthy behaviors.

2. Fagerlin et al. risk-communication principles â *Med Decis Making* 2007

Why: Absolute risks with denominators (e.g., â 2 in 100â) outperform relative-only framing for comprehension and appropriate worry.

In the dialogue: Opening script and system prompt require absolute risk + age-matched comparator; forbid â double your riskâ without absolute numbers. In the implemented prototype, the nci-natural-frequency-001 evidence chunk (National Cancer Institute â see EVIDENCE_REGISTER.md) grounds this natural-frequency framing.

3. Motivational Interviewing (OARS) â Rollnick, Miller & Butler, 2008

Why: Direct persuasion increases resistance in health contexts; MI elicits change talk.

In the dialogue: Elevated branch ends with open questions (â What would help you take the next step?â); chat suggestions are user-paced; assistant reflects rather than commands.

Avatar choice: â Mayaâ â warm female health educator

Selection criteria (research-informed):

- Trust & warmth: Health messages from perceived-credible, empathetic presenters increase recall and reduce defensive processing (source: health communication meta-literature; ACS/NCI patient-education tone).
- Role clarity: Introduces self as AI health educator, not a clinicianâ reduces authority